



Front-Yard Oak Savanna Edge

A Minnesota-native pollinator garden, designed and to be built

SITE

8547 Crane Dance Trail · Eden Prairie, MN 55344

Lot 0.36 ac · Hennepin County · USDA 4b/5a microclimate

DESIGNED FOR

Arun & Rima Batchu

DESIGNER · PACKAGE

Claude (acting as MN native-plants specialist) · v1.0

DESIGN PACKAGE

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Print 2-sided. Fold a corner to bookmark where you are. Write in the notes pages.

01 — VISION

A small pocket of Big Woods savanna

We're replacing the declining mugo pine bed in front of the south-facing bay window with a Minnesota-native pollinator garden in the Oak Savanna Edge style — a pocket of the tallgrass prairie / oak savanna community that grew on this Hennepin County moraine before settlement.

Anchored at the west end by a Serviceberry whose white April bloom echoes the entry crabapple's pink, flanked by Ninebark shrubs pruned low to preserve the indoor sightline through the bay window, with drifts of native prairie forbs in warm savanna tones — Wild Bergamot, Purple Coneflower, Butterfly Milkweed, Wild Lupine, False Blue Indigo, Smooth Blue Aster.

The current river-rock drainage swale becomes a fully-vegetated bioswale (no rock — Rima's preference) where native sedges and rushes form a fibrous root mat that armors the flow path while looking like a meadow.

A meandering limestone path enters from the existing east concrete walkway and arrives at a partially-shaded bench pocket in the west third — a quiet retreat tucked under the Serviceberry, oriented to look back through the bloom toward the entry crabapple and cul-de-sac borrowed view.

02 — SITE FACTS

What the property tells us

Lot 0.36 acres (~15,700 sq ft). Single-family home built 1997, 5,500 sq ft. Tan stucco with wood-shake roof. Parcel 1311622340067 in Hennepin County.

South-facing front bed, full sun (>6 hours daily). The stucco wall radiates heat — effectively raising the bed's hardness by half a USDA zone (zone 5a microclimate). Drought-tolerant prairie species thrive here.

Soil is almost certainly Hayden, Lester, or Kilkenny loam — the well-drained loamy glacial till of the Hennepin moraine. Confirm via the USDA Web Soil Survey for this parcel before the installer breaks ground. No amendment is likely needed.

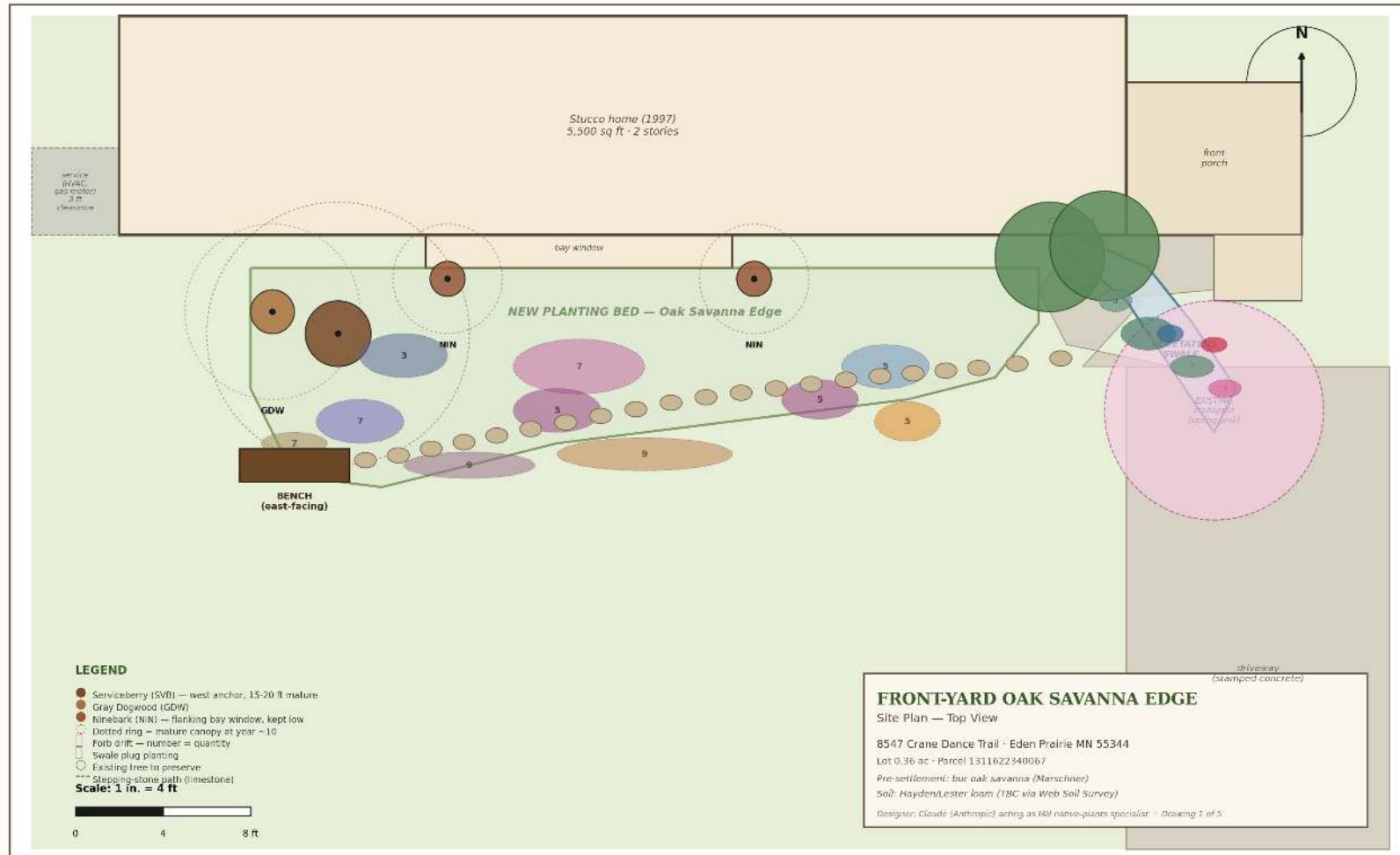
Pre-settlement vegetation (Marschner map): Big Woods / prairie-forest border subsection. Bur oak savanna on rolling moraine ridges, tallgrass prairie on flatter ground, maple-basswood forest in fire-protected ravines. The Oak Savanna Edge design is not a stylistic choice — it's the literal reconstruction of what grew on this land before 1850.

Existing assets to preserve: two mature arborvitae anchor the east end of the bed; a flowering crabapple frames the front entry in spring; a boxwood-yew hedge wraps the front porch foundation; the bay window is the family's primary indoor view — looking south across the lawn to a borrowed view of a flowering tree on the cul-de-sac.

Notes

Walk the bed before the installer arrives. Note: exact bed length, downspout location, where snow piles in winter, where the hose reaches.

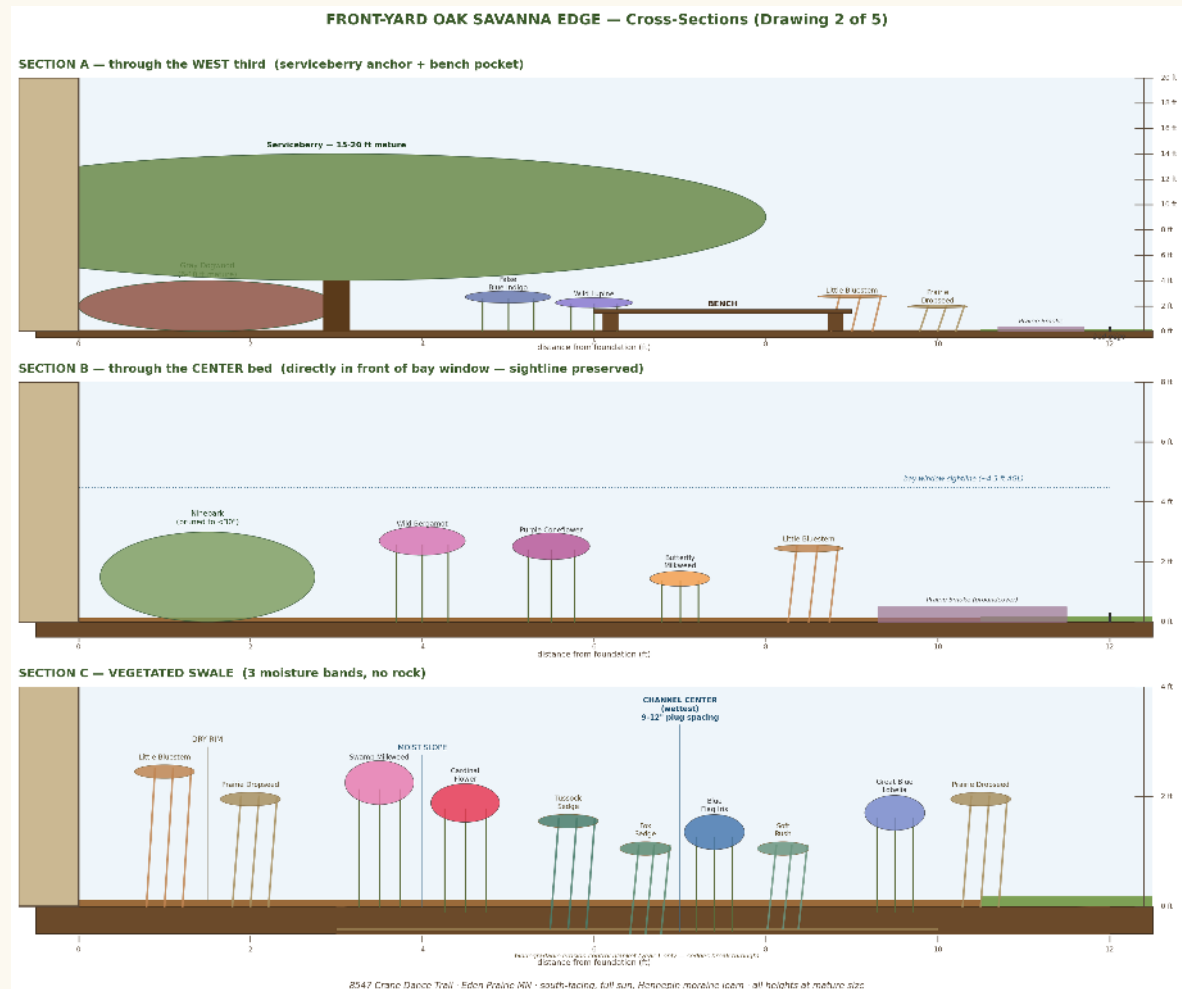
Top-down master drawing



Scale 1" = 4 ft. North up. Curved bed expands forward into the lawn, deepest at the west third where the bench pocket lives. The vegetated swale (steel blue) replaces the river-rock channel.

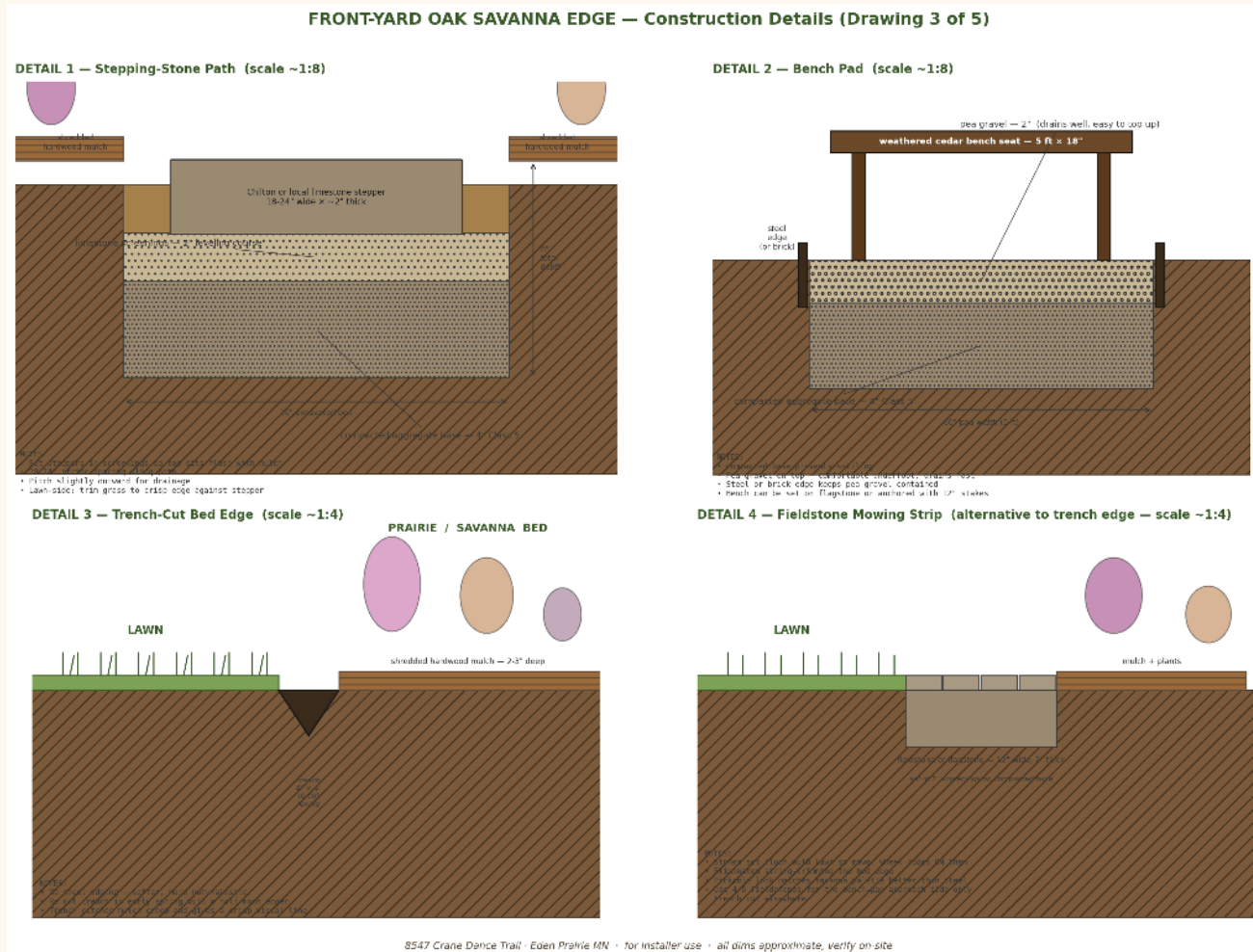
04 — CROSS-SECTIONS

Side views, three cuts



A: through the west third — Serviceberry over bench, tiered prairie forbs out to the lawn edge. B: through the center bed — Ninebarks pruned LOW so the bay-window sightline at ~4.5 ft AGL stays open. C: through the vegetated swale — three moisture bands; sedges and rushes armor the channel; biodegradable blanket year 1.

Path, bench pad, edges



Detail 1: stepping-stone bed prep — 4" Class 5 base, 2" screenings. Detail 2: bench pad on pea gravel. Detail 3: trench-cut edge — soft, naturalistic, re-cut each spring. Detail 4: fieldstone mowing strip as a low-maintenance alternative.

Notes

Installer feedback. Capture: do they have savanna-palette experience? what's their warranty on woody plants? do they handle vegetated bioswales (not just dry creeks)?

06 — PLANTING PLAN

Every plant, every quantity

BACKBONE — woody anchors

#	COMMON NAME	SCIENTIFIC	QTY	MATURE
1	Serviceberry	Amelanchier laevis	1	15-20 ft
2-3	Ninebark	Physocarpus opulifolius	2	5-8 ft *
4	Gray Dogwood	Cornus racemosa	1	6-10 ft

* prune ninebarks below 30" to preserve bay-window sightline

FORBS & GRASSES — main bed

#	COMMON NAME	SCIENTIFIC	QTY	BLOOM
5	False Blue Indigo	Baptisia australis	3	May-Jun
6	Wild Bergamot	Monarda fistulosa	7	Jul
7	Smooth Blue Aster	Symphyotrichum laeve	5	Sep-Oct
8	Wild Lupine	Lupinus perennis	7	May-Jun
9	Purple Coneflower	Echinacea purpurea	10	Jul-Sep
10	Butterfly Milkweed	Asclepias tuberosa	5	Jun-Aug
11	Little Bluestem	Schizachyrium scoparium	9	Sep+
12	Prairie Smoke	Geum triflorum	9	May
13	Prairie Dropseed	Sporobolus heterolepis	7	Aug

Planting Plan, continued

VEGETATED SWALE — three moisture bands

	COMMON NAME	SCIENTIFIC	QTY	BAND
14	Blue Flag Iris	Iris versicolor	3	wet
15	Tussock Sedge	Carex stricta	9	channel
·	Fox Sedge	Carex vulpinoidea	7	channel
·	Soft Rush	Juncus effusus	5	channel
·	Swamp Milkweed	Asclepias incarnata	3	moist
·	Cardinal Flower	Lobelia cardinalis	3	moist
·	Great Blue Lobelia	Lobelia siphilitica	3	moist

SERVICE-AREA GROUNDCOVER — west wall buffer

	COMMON NAME	SCIENTIFIC	QTY	USE
·	Pussytoes	Antennaria neglecta	7	stepable
·	Wild Strawberry	Fragaria virginiana	5	stepable

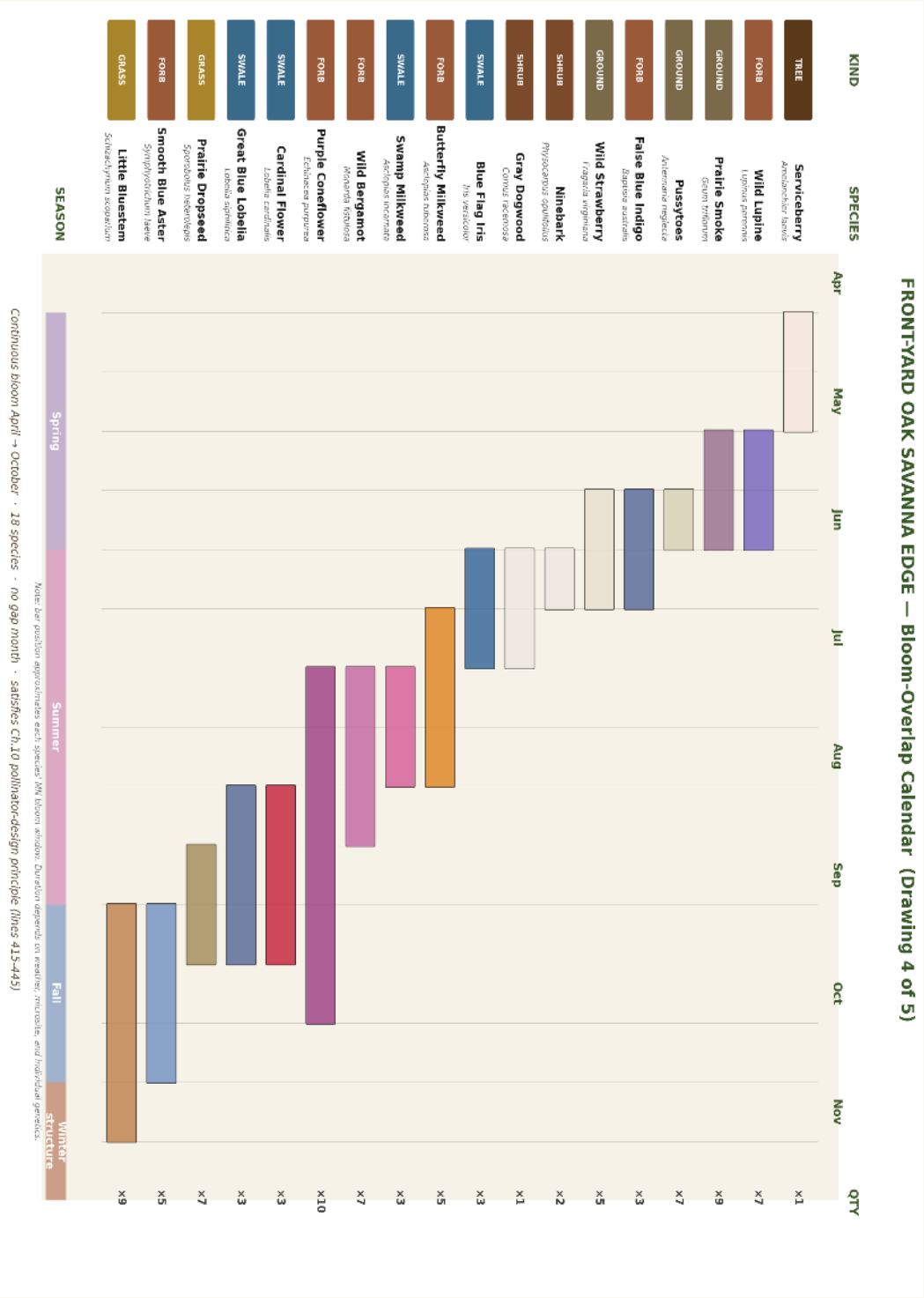
Plant total: ~111 specimens

Includes 4 woody backbone, 62 forbs & grasses, 33 swale plugs and 1-gal, 12 service-area groundcover.

Recommended sources

- Prairie Moon Nursery — Winona MN (mail order plugs & seed)
- Outback Nursery — Hastings MN (1-gal natives, shrubs, trees)
- Glacial Ridge Growers — Glenwood MN (wholesale plugs)
- Landscape Alternatives — Shafer MN (design + supply)

Continuous bloom April → November



Rotate the booklet 90° clockwise to read. Months run top to bottom, species along the right edge. 18 species, no gap month.

Notes

Plants to add or substitute. What's blooming in your favorite garden right now that you want here too? Note any species you saw at the Arboretum and want included.

The day after planting



Honest expectation-setting. Visible mulch and bare soil between plants. Plants installed at 1-quart and 1-gal container size. Most forbs won't bloom heavily this first season — root establishment.

09 — YEAR 3

Established



Drifts have filled in. Most forbs bloom on schedule. Pollinator populations have discovered the garden. Bluestem and Dropseed are at three-quarters mature size.

Matured



Late-afternoon August. Cardinal Flower in red bloom. Ninebarks at full height; Serviceberry casting dappled shade. The look is wild and intentional — clearly designed but no longer maintained-looking. Self-seeded Black-eyed Susan has filled small gaps.

09B — YEAR 3, MAY

The spring wave



Mid-May. Wild Lupine in violet-blue, Prairie Smoke groundcover in dusty pink, Baptisia just emerging. Serviceberry leafing out. Ninebarks budding. The vegetated swale is filling with fresh sedge growth. Different mood, same garden, three months earlier.

What the seating pocket feels like



Closer view of the west-third bench pocket. Weathered cedar bench under the Serviceberry, surrounded on three sides by Wild Bergamot and Cardinal Flower. Stepping-stone path entering from the right. Late afternoon golden side light. This is the room you sit in.

12 — MOOD BOARD

Walk these gardens first

Browse these with Rima before signing the install contract. The point is to absorb the visual vocabulary so you recognize the look as it grows in.

VISIT IN PERSON — THREE TWIN CITIES REFERENCES

- **Minnesota Landscape Arboretum — Prairie Garden**
Chaska, 25 min from Eden Prairie. Single best visit.
- **Eloise Butler Wildflower Garden**
Theodore Wirth Park, Mpls. Mature woodland-edge.
- **Lebanon Hills Regional Park — savanna restoration**
Eagan. The historic community of your land.

SPRING

- **Serviceberry in April bloom + Prairie Smoke**
search Wikimedia Commons: Amelanchier laevis, Geum triflorum
- **Wild Lupine in May**
Lupinus perennis · the iconic blue wave

SUMMER

- **Wild Bergamot + Purple Coneflower at peak**
Monarda fistulosa, Echinacea purpurea · bee storm
- **Butterfly Milkweed with Monarchs**
Asclepias tuberosa · vivid orange umbel

FALL + WINTER

- **Smooth Blue Aster + Cardinal Flower**
late-season blue and red
- **Little Bluestem — copper season**
Schizachyrium scoparium · best winter structure

Full URL list lives in mood-board.md in the design package folder.

How to keep it alive

First-year care

WATERING

Every 5-7 days for first 6 weeks; every 7-10 days through first growing season; only during drought after that.

EROSION BLANKET

Biodegradable jute or coir over the swale channel for year 1. Sedges break through it.

NO FERTILIZER

Native prairie plants are adapted to lean soils; fertilizer fuels weeds and makes forbs flop.

WEED PULL

Every 2 weeks first 2 summers. Creeping charlie regrowth is the main offender.

Year-by-year evolution

YEAR 1 · Looks 'planted'. Visible bare soil. Many forbs don't bloom heavily — root establishment year.

YEAR 2 · Drifts begin to fill in. Most forbs bloom. Sedges close the swale channel.

YEAR 3 · Established look. Bluestem at full size. Pollinator populations have discovered the garden.

YEAR 5+ · Self-managing meadow. Some species self-seed where they're happy. Edit gently.

Risk register

- **Storm scour through swale year 1**
erosion blanket; consider French drain
- **Creeping charlie reinfestation**
hand-pull pre-removal; spot-treat year 1
- **Lupine establishment failure**
MN-provenance plugs from Prairie Moon; or fall seed
- **Serviceberry transplant shock**
install in fall; deep-water through year 1
- **Snowplow crushes path edge**
stake path through first winter

Notes

Site walk-through. Things you noticed standing in the bed.

Notes

After install — first-month observations. What thrived. What looked rough. What surprised you.

15 — PLANT PROFILES

The Sixteen Plants

Botanical plates with qualities, care, history, and reason to be

The pages that follow profile every species in the design, from the Serviceberry that anchors the west end to the Tussock Sedge that armors the swale channel. Each plate is a hand-painted-style botanical illustration in the spirit of Curtis's Botanical Magazine, generated specifically for this project. Each profile carries four short sections: what the plant looks like, how to care for it, where it comes from, and the ecological reason it earns its place in your garden.

— Bree the Bee approves —

Serviceberry

Amelanchier laevis

Allegheny Shadbush · Juneberry



QUALITIES

A small multi-stem tree or large shrub of the rose family, growing 15 to 25 feet at maturity with a graceful rounded crown. Smooth pale-gray bark, and oval finely-toothed leaves that emerge bronze-purple, mature to soft green, and turn brilliant orange-red in fall. White five-petaled flowers appear in drooping clusters before the leaves in mid-April — one of the earliest woody bloomers in the Minnesota landscape — followed in June by dark purple edible berries that birds strip in a few days.

CARE

Plant in fall or early spring as a 6 to 8 ft balled-and-burlapped specimen, not bare-root. Full sun to part shade; tolerates loam, clay, or sandy soils so long as drainage is reasonable. Drought-tolerant once established. No regular pruning required; remove the occasional sucker to maintain a single-trunk or open multi-stem form. Hardy to USDA Zone 3. Avoid waterlogged sites and overly rich soils — they encourage soft growth that's prone to fire blight.

HISTORY

The name "serviceberry" descends from "sorbus," the Latin name for a related fruit. To the Anishinaabe and Dakota of Minnesota this tree is "Juneberry" — its fruits, eaten raw or pounded with meat and fat into pemmican, were among the first fresh foods of the year. Lewis & Clark recorded the species as a key wild food of the Great Plains, and it has been in continuous cultivation since.

REASON TO BE

A serviceberry is a small ecological universe. April flowers feed early-emerging native bees when almost nothing else is open. June berries feed Cedar Waxwings, Robins, Catbirds, and dozens of other songbirds. Its leaves host the larvae of more than a hundred native moth and butterfly species. In a designed garden it offers four seasons of structure — spring bloom, summer fruit, fiery fall color, winter silhouette. Plant it as the anchor and the rest of the garden gathers around it.

Ninebark

Physocarpus opulifolius

Eastern Ninebark



QUALITIES

An arching deciduous shrub of stream-banks and rocky openings, growing 5 to 8 feet tall and as wide. The common name comes from its habit of peeling bark in long curling strips, exposing layer after layer of red-brown to cinnamon-colored inner bark — a landmark in the winter garden. Three-lobed maple-like leaves and clusters of small white flowers in late June give way to reddish papery seed capsules that persist into fall.

CARE

Tough as nails. Full sun to part shade; tolerates clay, sand, drought, occasional flooding, urban pollution, road salt, and rabbits. Hardy to USDA Zone 2. Plant 1-gallon or 3-gallon container stock; spaces about 5 ft apart for a loose hedge. Prune in late winter — cut a third of oldest stems to ground each year to keep the shrub vigorous and the bark display visible. Avoid 'Summer Wine' and similar purple-foliage cultivars in stucco-architecture settings — the foliage clashes.

HISTORY

Ninebark's name appears across Indigenous, Colonial, and modern horticulture. The Cherokee used a bark decoction as an emetic and for women's medicine; the Ojibwe used it in post-childbirth care. Settlers in eastern North America named it "ninebark" for the apparent nine layers of shedding bark. It became a foundational shrub in 20th-century American landscape design before the cultivar explosion of the 2000s; the straight species is now experiencing a quiet revival among ecological designers.

REASON TO BE

Ninebark anchors residential bed design with a visual presence that shifts every month: white pollinator-loaded flowers in June, reddish seedheads through summer, golden leaves in October, and the celebrated peeling bark from November through April. Native bees, hover flies, and small butterflies work the bloom. Birds nest in the dense branching. It is one of the few native shrubs that holds its own visually against architectural elements — stucco, stone, brick — without disappearing into the green.

Gray Dogwood

Cornus racemosa

Northern Swamp Dogwood



QUALITIES

A clonal native shrub forming dense colonies 6 to 10 feet tall and equally wide. Slender gray-brown stems hold narrow oval leaves that turn dusty purple-red in autumn. Flat-topped clusters of small white flowers in early summer mature to chalk-white berries on bright red pedicels — one of the most striking fall combinations in the Minnesota landscape. Birds typically strip the fruit within a week or two; the red pedicels persist into winter as decorative accents.

CARE

Adaptable to nearly any soil except deep sand or saturated wetland. Full sun gives best fruit set; tolerates part shade. Suckers slowly to form thickets — a feature, not a bug, in informal native plantings, but in formal beds you may want a root barrier or annual sucker pruning. Hardy to USDA Zone 3. Disease- and pest-free. No regular feeding or watering needed once established.

HISTORY

Gray dogwood was used by several Indigenous peoples for tool handles, basket rims, and arrows; the straight, flexible young stems take a clean shave. Colonial herbalists pressed bark and root for a quinine substitute (it's a relative of the medicinal *Cornus florida*). In Minnesota's prairie-forest border ecosystem, it filled the role of "thicket maker" — providing the dense shrubbery that separated savanna openings from closed woodland.

REASON TO BE

Gray dogwood is the workhorse of bird-supporting shrub plantings. Its berries feed at least 98 documented bird species, including Brown Thrashers, Wood Thrushes, Catbirds, and migrating Warblers. The dense thicket form provides nesting cover for ground-nesting and shrub-nesting birds. As a fast-establishing native, it serves as a placeholder while slower-growing trees like oak mature; in larger restoration projects it forms the ecological scaffolding under which an oak savanna or edge habitat assembles itself.

False Blue Indigo

Baptisia australis

Wild Blue Indigo · Blue False Indigo



QUALITIES

A long-lived herbaceous perennial of the pea family, growing 3 to 4 feet tall with the architectural presence of a small shrub. Blue-green clover-like leaves on branching upright stems form a tidy mounded clump. In late May and June, vertical racemes of indigo-blue pea-like flowers rise above the foliage, followed by inflated black seed pods that rattle decoratively into winter. The deep taproot makes Baptisia nearly impossible to transplant once established — and nearly impossible to kill.

CARE

Plant from 1-gallon container in spring, into open ground where it will live for the rest of your gardening life. Full sun, average to dry soil; tolerates drought, lean soil, and clay. Slow to establish — expect a quiet first two summers — then bullet-proof. Hardy to USDA Zone 3. As a nitrogen-fixing legume, it actively improves soil for neighboring forbs. Avoid moving once mature; if you must, do it in early spring with a tarp and prepare to lose half the root system.

HISTORY

The genus name *Baptisia* derives from Greek *baptein*, "to dye" — Indigenous peoples used several Baptisia species as a substitute for the imported indigo dye, though the color is muted compared to true *Indigofera*. Cherokee and Choctaw used a root preparation for toothache and as a purgative. In 19th-century horticulture Baptisia was promoted as a substitute for the southern lupines, since it is more cold-hardy. It has remained a stalwart of MN prairie restoration ever since.

REASON TO BE

Baptisia anchors a prairie or savanna planting in time. While neighboring forbs cycle through five-year boom-and-decline rhythms, Baptisia is in year fifteen still in the same spot, getting better. Its early-summer bloom feeds long-tongued bumblebees and the Wild Indigo Duskywing skipper, which uses Baptisia as its larval host plant. The seed pods are an ornamental feature into winter. As a nitrogen fixer, it conditions soil for the rest of the planting. Plant it as the structural backbone.

Prairie Smoke

Geum triflorum

Old Man's Whiskers · Three-flowered Avens



QUALITIES

A low evergreen rosette of fern-like compound leaves, 6 to 10 inches tall in flower, that hugs the ground year-round. In May, slender stems each bearing three nodding pink-purple urn-shaped flowers rise above the foliage. As the flowers age the petals fall and the styles elongate into long feathery plumes — pink, smoky, twisting — that give the plant its name. The plumes catch the light and the wind for weeks. By midsummer the seedheads dry and disperse, leaving the evergreen rosette behind.

CARE

Full sun, well-drained soil; ideal for sandy or rocky sites. Hates wet feet — site away from downspouts and low spots. Plant from 4-inch plug or quart container in spring or fall, 9 to 12 inches apart for groundcover. Hardy to USDA Zone 3. Slow to spread but extremely long-lived once established — a clump can persist for decades. Avoid mulching directly over the rosette; the evergreen leaves want air.

HISTORY

Prairie smoke is one of the iconic remnant-prairie indicator species in Minnesota — finding a colony in a ditch or pasture is considered evidence of pre-settlement soil. The Blackfeet used a root decoction as eyewash and as a tea for stomach upset. The seed plumes were occasionally used for arrow fletching and as kindling. The name "old man's whiskers" appears across Indigenous and settler vocabularies for the same reason — the feathery plumes look exactly like grizzled hair.

REASON TO BE

Prairie smoke is the path-edge groundcover for which every native garden secretly hopes. It blooms before almost everything else in the prairie, drawing early queen bumblebees out of dormancy. Its evergreen winter presence keeps the bed from looking dead in March and October. The seedplumes are unforgettable — visiting neighbors stop in their tracks. Plant it where children can touch the plumes; that tactile encounter is how prairie ecology becomes real for a 6-year-old.

Wild Bergamot

Monarda fistulosa

Bee Balm · Horsemint



QUALITIES

A clumping mint-family perennial 2 to 4 feet tall, with square stems, opposite gray-green leaves that smell sharply of oregano when bruised, and rounded two-inch-wide pom-poms of lavender-pink tubular flowers in July. Flowers persist three to four weeks; the structural seedheads remain handsome through fall and into winter. Forms a slowly-expanding colony by underground rhizomes.

CARE

Full sun is best; tolerates light shade. Average to dry soil; drought-tolerant. Deer-resistant. In humid climates or crowded plantings, watch for powdery mildew on lower leaves; thinning the clump and improving airflow usually fixes it without spraying. Hardy to USDA Zone 3. Cut back to 6 inches in late winter; the hollow stems (*fistulosa* means "hollow") house overwintering native bees, so don't tidy too early in spring.

HISTORY

Wild bergamot is the wild ancestor of cultivated Bee Balm and is named for its similarity in scent to bergamot orange (*Citrus bergamia*) — Earl Grey tea's flavor note. The Ojibwe and Menominee used the leaves as a medicinal tea for respiratory illness, and the species is the primary source of thymol in pre-aspirin North American medicine. Following the Boston Tea Party, colonists used wild bergamot as a black-tea substitute ("Oswego tea").

REASON TO BE

If you plant only one pollinator forb, plant this. Wild bergamot draws an extraordinary diversity of visitors: long-tongued native bumblebees, Hummingbird Clearwing moths, Ruby-throated Hummingbirds, Eastern Tiger Swallowtails, and dozens of native bee species. Its hollow stems shelter overwintering tunnel-nesting bees. It is a workhorse that delivers ecological value at every stage — bloom, seed, winter cover.

Purple Coneflower

Echinacea purpurea

Eastern Purple Coneflower



QUALITIES

A clumping prairie composite 2 to 3 feet tall, with rough hairy lance-shaped leaves and large daisy-like flowers from July through September. The drooping rose-pink ray petals surround a spiny copper-orange central cone made of hundreds of tiny disc florets. Cones persist into winter as architectural seedheads favored by goldfinches.

CARE

Full sun, average to dry well-drained soil. Tolerates clay better than most coneflowers. Hardy to USDA Zone 3. Plant from 1-gallon container or seed in fall; self-seeds modestly into receptive areas (a feature in naturalistic plantings). Deadheading extends bloom but removes the seed source for goldfinches; consider leaving half. Cut stems to 6 inches in late winter.

HISTORY

Echinacea derives from Greek *echinos*, "hedgehog" — for the spiny central cone. Plains tribes used the root extensively as an immune-system modulator, applying it for everything from snakebite to toothache to colds. European settlers adopted the use, and a 19th-century Nebraska physician's commercial Echinacea preparations became one of the most widely-sold patent medicines of the era. Modern research has confirmed several of the traditional uses; Echinacea remains a top-selling herbal supplement worldwide.

REASON TO BE

Purple coneflower is the public face of native-plant gardening: instantly recognizable, beloved by visitors, tolerant of mistakes, and ecologically generous. Long-tongued bees and butterflies (especially Monarchs, Fritillaries, and Painted Ladies) work the cones; goldfinches and other small finches harvest the seeds through fall and winter. In a residential design it carries the show from July through September while neighboring forbs hand off bloom.

Smooth Blue Aster

Symphyotrichum laeve

Smooth Aster



QUALITIES

A graceful prairie aster 2 to 3 feet tall, with smooth blue-green clasp leaves on slender arching stems. From September into October it produces a cloud of inch-wide sky-blue daisies with yellow centers, often the last major bloom in the Minnesota prairie before frost. The smooth foliage and tidy form distinguish it from many of its weedier aster cousins.

CARE

Full sun is best; tolerates light shade. Average to dry soil; tolerates drought, clay, and lean conditions. Deer-resistant. Hardy to USDA Zone 3. Pinch back stems in early June to encourage compact branching and reduce any tendency to flop. Self-seeds modestly. Long-lived and clump-forming rather than aggressively spreading.

HISTORY

Asters were among the most diverse and difficult plant groups for early North American botanists; the genus has been split, lumped, and renamed multiple times — the modern *Symphyotrichum* is a 1995 reorganization of the old North American "asters." Indigenous peoples used various asters in smoke for hunting medicine and as a diaphoretic tea. Smooth blue aster specifically appears across the prairie-forest border subsection of Minnesota's pre-settlement vegetation map.

REASON TO BE

Smooth blue aster is the closing act of the prairie season — when bergamot and coneflower are setting seed, this aster lights up the bed in cool sky-blue. It is a critical late-season nectar source for migrating Monarchs and queen bumblebees fattening for hibernation. The Pearl Crescent butterfly uses it as a larval host. Its reliable upright form and rich blue color make it the easiest aster to slot into formal designs without the wild-edged look of New England aster.

Wild Lupine

Lupinus perennis

Sundial Lupine



QUALITIES

An iconic spring prairie forb 1 to 2 feet tall, with palmate leaves of 7 to 11 narrow leaflets that orient throughout the day to track the sun (hence "sundial lupine"). In May and June, vertical racemes of violet-blue pea-like flowers create the legendary wave of color that once defined oak savannas across the upper Midwest. Goes dormant in midsummer, disappearing entirely by August.

CARE

Plant in spring or fall as plug or 1-gallon stock — MN-genotype provenance is essential. Full sun; sandy or well-drained loamy soil. Lupine refuses heavy clay and wet sites. Slow to establish; expect a quiet first year. Hardy to USDA Zone 3. As a legume, fixes nitrogen and conditions soil. Do not amend or fertilize — rich soil makes lupine flop and shortens its life.

HISTORY

Wild lupine is the **only** food plant of the federally endangered Karner Blue butterfly (**Plebejus melissa samuelis**). The Karner Blue was widely abundant across Minnesota's pre-settlement oak savannas. As savannas were converted to agriculture and shaded out by fire suppression, lupine populations crashed and the Karner Blue followed; the butterfly was extirpated from Minnesota in 2008 and listed as endangered nationally. Reintroduction efforts depend on restoring lupine.

REASON TO BE

To plant wild lupine in Minnesota in 2026 is to participate in the restoration of an entire vanished ecosystem. Even one residential planting matters: it feeds bumblebees and a host of native specialist bees that no other species supports. If Karner Blues are ever reintroduced to your area (the work is active in Wisconsin and may extend to MN), your lupine drift will be a stepping-stone. Its ephemeral spring presence — blooming gloriously then fading — teaches a lesson about prairie phenology that nothing else can.

Butterfly Milkweed

Asclepias tuberosa

Butterfly Weed · Pleurisy Root



QUALITIES

A clumping prairie milkweed 1 to 2 feet tall, with narrow alternate leaves on hairy upright stems. From June into August it produces vibrant clusters of five-pointed orange flowers — the most saturated orange in the native flora — followed by typical milkweed spindle-shaped pods that split to release silk-tufted seeds. Unlike most milkweeds it does *not* exude white latex when broken — the sap is clear, hence the genus name *tuberosa* meaning "with tuberous root."

CARE

Full sun, well-drained soil; ideal for sand or sandy loam. Drought-tolerant. Deer-resistant. The deep taproot makes Butterfly Milkweed nearly impossible to transplant — plant once, leave alone. Hardy to USDA Zone 3. Slow to emerge in spring (mark its location with a stake the first year so you don't accidentally weed it). Long-lived once established.

HISTORY

Butterfly milkweed's other common name, "pleurisy root," reflects its widespread historical use among Indigenous peoples and 19th-century herbalists for respiratory illness — the Choctaw, Omaha, Ponca, and Dakota used a root preparation. The bright orange dye from the flowers and stems was used in basketwork. European observers noted the plant as one of the most visually striking of North American natives, and it has been in cultivation since the 1690s.

REASON TO BE

Butterfly milkweed is one of three *Asclepias* species that serve as primary host plants for the Monarch butterfly — without milkweed, no Monarchs. Its summer bloom is also a magnet for Eastern Tiger Swallowtails, Fritillaries, and dozens of native bee species. The saturated orange is so visually striking it provides the design's color punctuation. Plant in drifts of 5 or more for visual mass and pollinator-attraction amplification.

Little Bluestem

Schizachyrium scoparium

Beard Grass



QUALITIES

The signature native warm-season grass of dry prairies and oak savannas, growing 2 to 3 feet tall in a fountain-shaped clump. Blue-green summer foliage shifts in autumn to the famous copper-and-mahogany tones that define Minnesota prairie color through October. Feathery silver seedheads catch backlighting from late summer through winter — among the best sources of structural light in any native garden.

CARE

Full sun, well-drained soil; sand to loam. Drought-tolerant once established. Deer- and rabbit-resistant. Hardy to USDA Zone 3. Plant from 1-gallon container 18 to 24 inches apart for a meadow matrix; closer for a tighter ribbon. Leave standing through winter — the structure is the point. Cut to 6 inches in late February or March before new growth.

HISTORY

Little bluestem co-dominated the tallgrass prairie with Big Bluestem (**Andropogon gerardii**) — together they formed the iconic 6-foot grass sea of pre-settlement Minnesota. The deep root systems (down to 6 feet) built the legendary prairie soils that drew settlers and destroyed the prairies in equal measure. Today, less than 2% of Minnesota's tallgrass prairie remains. Each residential planting of little bluestem is a small act of restoration.

REASON TO BE

Little bluestem is the structural connective tissue of a savanna-style residential bed. Planted in repeated drifts, it weaves between forb mass and unifies the composition. It provides cover for ground-nesting native bees and small birds; its seedheads feed sparrows in winter. It hosts the larvae of many native butterflies and skippers. And in November when every forb has gone brown, the copper bluestem still glows like an ember through the snow.

Prairie Dropseed

Sporobolus heterolepis

Northern Prairie Dropseed



QUALITIES

A fine-textured warm-season bunchgrass forming a graceful arching mound 2 to 3 feet tall and equally wide. Hair-thin medium-green leaves give it the softest texture of any prairie grass. Airy panicles of tiny seeds appear in August, releasing a mild buttered-popcorn or coriander fragrance that surprises first-time visitors. Foliage turns golden bronze in fall.

CARE

Full sun is essential; tolerates almost any soil including clay. Drought-tolerant once established but appreciates occasional summer water. Hardy to USDA Zone 3. Slow to establish (3-year root development); plant from 1-gallon stock for fastest results. Long-lived once settled — an undisturbed clump can persist for decades. Cut to 6 inches in late winter.

HISTORY

Prairie dropseed was a co-dominant grass of the drier-mesic prairies of southern Minnesota and was a Dakota food plant — the seeds, parched and ground, were used as flour. Following the conversion of prairies to agriculture, dropseed retreated to remnant sites; it is now a flagship indicator species for prairie restoration quality. Its slow establishment and refusal to transplant easily make it a marker of patient gardeners and undisturbed soil.

REASON TO BE

Prairie dropseed is the front-edge grass of choice for any bed that meets a path or a lawn. Its fountain form softens edges; its fine texture contrasts beautifully with broad-leaved forbs; and the late-summer popcorn scent is one of the small joys of native gardening that no garden-design book ever mentions. Native bees and skippers use the foliage; goldfinches harvest the seed.

Blue Flag Iris

Iris versicolor

Northern Blue Flag · Larger Blue Flag



QUALITIES

A bold wetland iris 2 to 3 feet tall, with sword-shaped blue-green leaves rising from spreading rhizomes. In late May and June it produces large violet-blue flowers with bright yellow signal patches and dramatic dark veining on each falling petal — the classic iris form, scaled up. Forms colonies in damp soil over time. Rhizomes are toxic if eaten; do not confuse with edible Sweet Flag (**Acorus calamus**).

CARE

Best in moist to wet soil; thrives at the edge of a rain garden, vegetated swale, or pond. Tolerates average garden soil if watered. Full sun to part shade. Hardy to USDA Zone 3. Plant from 1-gallon stock or bare-root rhizome in spring or fall. Divide every 4-5 years to maintain vigor. Slugs occasionally chew young leaves; rarely a serious issue.

HISTORY

Blue flag was widely used among Indigenous peoples of the Great Lakes region, including the Ojibwe and Anishinaabe, as a powerful medicinal — but specifically for external use; the rhizomes contain compounds that are emetic and dermatologically active. The plant appears in ceremonial contexts in some Algonquian traditions. The name *versicolor* ("various-colored") refers to the elaborate yellow-and-purple signal patterning on the falls.

REASON TO BE

Blue flag iris is the visual anchor of a vegetated bioswale or rain garden. Its bold form and early-summer bloom announce the wet zone of a composition. Bumblebees and skipper butterflies work the flowers; the seed pods provide late-season structural interest. In a residential swale conversion (rock to vegetated), three or five clumps of Blue Flag elevate the channel from "grass meadow" to "designed wet garden" without losing any of the ecological function.

Tussock Sedge

Carex stricta

Upright Sedge



QUALITIES

A robust clumping sedge of fens, sedge meadows, and vegetated bioswales, forming distinct elevated "tussocks" 1 to 3 feet tall. Bright green narrow blades arch outward from a tightly clumped base; the tussock structure literally raises the plant slightly above the wet ground around it. Inconspicuous spring spikelets ripen to brown by midsummer. The fibrous root mat is the species' superpower — it armors flow paths in a way no other plant matches.

CARE

Plant from plug stock at 9 to 12 inches spacing in wet or moist soil — the closer spacing matters because the purpose is rapid canopy closure to outcompete weeds and armor against erosion. Tolerates seasonal flooding; tolerates partial drying once established. Full sun to part shade. Hardy to USDA Zone 3. No maintenance needed; can be selectively burned in a controlled spring burn.

HISTORY

Carex stricta was a dominant species of Minnesota's pre-settlement sedge meadows, which historically covered substantial acreage in poorly-drained lowlands across the state. Most sedge meadow has been tilled and converted to agriculture. The sedge meadow is one of the great "missing" Minnesota habitats — and tussock sedge is its keystone species. Each vegetated bioswale planted with **Carex stricta** is a small fragment of restored sedge meadow.

REASON TO BE

Tussock sedge is the structural species of a vegetated bioswale — the species that physically holds soil against stormwater flow with its dense fibrous root mat. Without sedge, a swale is just lawn or ditch; with sedge, it is a functioning piece of green stormwater infrastructure. The tussock form provides habitat for amphibians, ground-nesting birds, and specialist insects. As a substitute for a rock dry-creek it is an order of magnitude more ecologically valuable, and (after year 1) more visually beautiful.

Cardinal Flower

Lobelia cardinalis

Red Lobelia



QUALITIES

An upright wetland perennial 2 to 4 feet tall, with narrow lance-shaped leaves and a vertical spike of vivid scarlet two-lipped flowers from August into September. The red is so saturated it is almost shocking against late-summer prairie greens — the single most striking flower color in the Minnesota native flora. Short-lived (3-5 year) but reliably self-seeds in suitable conditions.

CARE

Moist to wet soil; the wetter the better. Full sun to part shade. Hardy to USDA Zone 3. Plant from 1-gallon stock in spring or fall; do not let the root ball dry out during establishment. Mulch lightly in fall to protect the basal rosette through winter. Allow self-seeding in compatible spots; replace as the original parent plants decline. Deer-resistant.

HISTORY

Named for its resemblance to a Cardinal's red robe, the plant has been cultivated since the 1620s. Indigenous peoples used the root for love medicine and for respiratory illness, though the alkaloids are toxic and the plant requires careful handling. Cardinal Flower appears in Audubon's earliest American botanical watercolors (1820s) — it has long been recognized as one of the most beautiful native plants of eastern North America.

REASON TO BE

Cardinal flower is *the* hummingbird plant. The Ruby-throated Hummingbird is its primary pollinator — the narrow tubular red flower is an evolutionary co-adaptation. In late August and September, when hummingbirds are migrating south and need high-energy nectar, a stand of cardinal flower is a refueling station. The vivid red also draws Eastern Tiger and Spicebush Swallowtails. In a vegetated bioswale, cardinal flower transforms the moist slope from "green stormwater function" to "emotional anchor."

Fox Sedge

Carex vulpinoidea

Brown Fox Sedge



QUALITIES

A clumping native sedge of wet meadows, ditches, and vegetated bioswales, growing 2 to 3 feet tall. Narrow bright-green blades arch outward from a tight basal clump. In June and July, distinctive bristly seedheads emerge — each one resembling a small foxtail or pipe-cleaner, which gives the species its name. The seedheads turn from green to golden brown as they ripen and persist into fall, providing structural interest after most prairie forbs have finished blooming.

CARE

Moist to wet soil; tolerates seasonal flooding and occasional summer drying once established. Full sun to part shade. Hardy to USDA Zone 3. Plant from plug stock at 9 to 12 inches spacing for fast canopy closure. As with all sedges, the close spacing matters because the purpose is to outcompete weeds and bind the soil with fibrous roots. No maintenance required; can be cut to 6 inches in late winter or burned on a controlled spring schedule.

HISTORY

The genus *Carex* contains over 2,000 species worldwide and is one of the largest plant genera; it includes some of the most ancient grass-like plants on Earth. Sedges were co-dominant species of the sedge meadows that once covered substantial acreage in poorly-drained Minnesota lowlands. The Ojibwe and Anishinaabe used several *Carex* species for woven mats and basketry. Most sedge meadow habitat in Minnesota has been drained and converted to agriculture; what remains is among the state's most endangered ecosystems.

REASON TO BE

Fox Sedge pairs with Tussock Sedge as the structural backbone of a vegetated bioswale — the species that physically holds soil against stormwater flow. The bristly seedheads feed sparrows and small finches through fall and winter. The species hosts the larvae of several skipper butterflies and Eyed Brown butterflies. In a residential bioswale conversion (rock to vegetated), Fox Sedge provides the visual texture and the structural function in equal measure.

Soft Rush

Juncus effusus

Common Rush



QUALITIES

A clumping wetland rush forming distinct fountain-like tussocks 2 to 3 feet tall. Smooth cylindrical bright-green stems rise vertically from a tight basal clump — the stems are the photosynthetic organs (true leaves are reduced to small basal sheaths). Inconspicuous brown panicles of tiny flowers burst out from the side of the stems near the top in early summer, persisting as structural seedheads through fall.

CARE

Wet to consistently moist soil; tolerates seasonal flooding. Full sun to part shade. Hardy to USDA Zone 3. Plant from plug stock or 1-gallon container, 12 to 18 inches apart for a clump-forming display. Long-lived; established clumps can persist for decades. No maintenance needed; can be cut to 6 inches in late winter, though leaving stems standing through winter provides cover for small wildlife.

HISTORY

Soft Rush is one of the few truly cosmopolitan plants — it grows wild on every continent except Antarctica. Throughout its range it has been used for centuries to weave chair seats, mats, and baskets. In medieval England, the pith of the stems was soaked in tallow to make "rushlights," the cheapest form of household lighting before commercial candles became affordable. The Anishinaabe and other Great Lakes peoples used it similarly for mat-making.

REASON TO BE

Soft Rush is a structural element of vegetated bioswales and rain gardens, providing vertical contrast to the horizontal arch of sedges and the showy spikes of Iris and Cardinal Flower. The dense clumps shelter amphibians, small birds, and overwintering native bees. It tolerates the seasonal flooding that defeats most other plants — making it a reliable choice for the low spots of a vegetated swale where standing water occasionally sits.

Great Blue Lobelia

Lobelia siphilitica

Blue Cardinal Flower



QUALITIES

An upright wetland perennial 2 to 3 feet tall, with lance-shaped leaves on stiff vertical stems. From August into September it produces dense terminal spikes of two-lipped sky-blue tubular flowers — the showiest true-blue native bloom of late summer in Minnesota. Each flower has a distinctive pair of white striped throats that guide pollinators to the nectar. Self-seeds modestly in suitable conditions.

CARE

Moist to wet soil; tolerates average garden soil if watered. Full sun to part shade. Hardy to USDA Zone 4. Plant from 1-gallon stock in spring; do not let the root ball dry out during establishment. Mulch lightly in fall. Short-lived (3-5 year) but reliably self-seeds. Companion-plant with Cardinal Flower; the two species occasionally hybridize naturally to produce striking purple-blooming offspring.

HISTORY

The species name *siphilitica* reflects its (now-debunked) historical use as a treatment for syphilis among Indigenous peoples; the root contains alkaloids that are toxic in large doses but were used medicinally in small preparations. The species has been cultivated since the 17th century, when it was introduced to European gardens as one of the first North American plant exports. The rich blue color was sought-after when blue flowers were rare in European horticulture.

REASON TO BE

Great Blue Lobelia provides a critical late-summer nectar source for long-tongued bumblebees, who are strong enough to pry open the two-lipped flowers; smaller bees can't. Its August-September bloom fills the gap between midsummer Bergamot and fall asters. Planted alongside Cardinal Flower in a vegetated bioswale's moist slope, the red-and-blue pairing is one of the most striking color contrasts in the native flora — and supports both bee and hummingbird pollinators in one composition.

Pussytoes

Antennaria neglecta

Field Pussytoes · Plantain-leaved Everlasting



QUALITIES

A low evergreen mat-forming perennial of dry prairies and oak openings. Silvery-gray rosettes of small spoon-shaped fuzzy leaves spread by surface runners to form a dense ground-hugging carpet 2 to 4 inches tall. In May, short 4-to-6-inch stalks rise above the foliage bearing tight white fuzzy flower clusters that look uncannily like a kitten's toepads — giving the species its common name.

CARE

Full sun, dry well-drained soil; ideal for sandy or rocky sites and sun-baked margins. Hates wet feet — site away from downspouts and drainage paths. Plant from plug stock 6 to 9 inches apart. Hardy to USDA Zone 3. Fully steppable once established — tolerates moderate foot traffic without damage, making it a rare native option for paths, stepping-stone gaps, and service-area buffers.

HISTORY

Pussytoes belong to a group of plants called "everlastings" because the dried flower clusters retain their form and color for years; they were widely used in Victorian dried-flower arrangements and memorial wreaths. Several Plains tribes used a leaf preparation as a remedy for snakebite (largely symbolic) and as a smoke for purification ceremonies. The genus is named *Antennaria* for the flower's resemblance to butterfly antennae.

REASON TO BE

Pussytoes is the larval host plant for the American Lady butterfly (*Vanessa virginiensis*) — caterpillars build silken nests on the silvery foliage. Few native groundcovers tolerate foot traffic, drought, AND host a charismatic butterfly all at once; this one does. In a residential design it solves the problem of "what to plant where service techs walk" — the HVAC buffer, the gas-meter approach — without resorting to non-native ground cover or sterile gravel.

Wild Strawberry

Fragaria virginiana

Virginia Strawberry · Mountain Strawberry



QUALITIES

A spreading native perennial groundcover 4 to 6 inches tall, with characteristic three-leaflet (trifoliate) compound leaves on slender stalks. White five-petaled flowers with bright yellow centers appear in May and June, followed by small red strawberries in June and July — true strawberries, smaller but more intensely flavored than commercial varieties. Spreads aggressively by surface runners (stolons) to form a dense knit-together mat.

CARE

Full sun to part shade; tolerates a wide range of soils from sandy loam to clay. Drought-tolerant once established. Hardy to USDA Zone 3. Plant from plug or 4-inch container 12 inches apart; will fill in within two seasons. Fully steppable. No pest or disease issues in residential settings. Removes runners as needed if you want to contain spread.

HISTORY

Fragaria virginiana is one of the two parent species of the modern cultivated strawberry (*Fragaria × ananassa*) — the cultivated berry is a hybrid produced in 1750s France between this North American species and the Chilean strawberry. The Anishinaabe called wild strawberry "ode'imín" (heart berry) and considered it sacred; June was traditionally Strawberry Moon (Ode'imín Giizis). The leaves have been used for tea since pre-contact times across most of the species' range.

REASON TO BE

Wild Strawberry is one of the few truly edible native groundcovers — the tiny berries that you (or the robins) harvest in June are real strawberries, concentrated in flavor. The flowers feed early-season native bees; the leaves host the larvae of the Grey Hairstreak butterfly. As a ground-hugging tough native, it converts the service-area buffer or any path edge from sterile mulch into ecologically functional habitat — and gives you something to taste in passing.

Swamp Milkweed

Asclepias incarnata

Pink Milkweed · Rose Milkweed



QUALITIES

An upright clumping milkweed 3 to 4 feet tall, with narrow lance-shaped leaves and rounded clusters of small mauve-pink star-shaped flowers from July through August. Unlike Common Milkweed it does not spread by rhizomes — it forms a tidy clump that holds its place indefinitely. Spindle-shaped pods split in fall to release the iconic silk-tufted milkweed seeds.

CARE

Best in moist to wet soil; tolerates average garden soil if watered. Full sun. Hardy to USDA Zone 3. Plant from 1-gallon stock; like all milkweeds, slow to emerge in spring (mark the location). Long-lived once established; clumps remain in place rather than spreading. Deer-resistant. The flowers attract considerable pollinator traffic — site near a viewing area.

HISTORY

All North American milkweeds were used variously by Indigenous peoples — the silky seed floss as fiber (stuffed into life jackets during WWII when imported kapok was unavailable), the flower buds and very young shoots as cooked food, the latex sap as a wound dressing. Swamp milkweed specifically appears in Anishinaabe and Dakota wetland-edge plant lists. The common name is misleading — it grows fine in average garden soil so long as moisture is reliable.

REASON TO BE

Swamp milkweed is the second of three primary Monarch host plants. Where Butterfly Milkweed offers drought-tolerant prairie habitat and Common Milkweed offers spreading colonies, Swamp Milkweed offers a well-behaved clumping habit perfect for residential design. Female Monarchs lay eggs on swamp milkweed throughout July and August; the leaves are the exclusive food of the caterpillars. In a vegetated bioswale's moist slope, three Swamp Milkweed clumps make the swale a Monarch nursery.



Restoring a small pocket of pre-1850 Hennepin County to its native hands.

This package is a living document. The plan is good. The plants are right for this site. The renderings are honest about year 1, year 3, year 5. The numbers are real. The mood board points to where to walk before the install — the Arboretum's prairie garden, Eloise Butler, Lebanon Hills.

Take it with you. Write in the margins. Come back to it when the Cardinal Flower blooms in August of year 3.